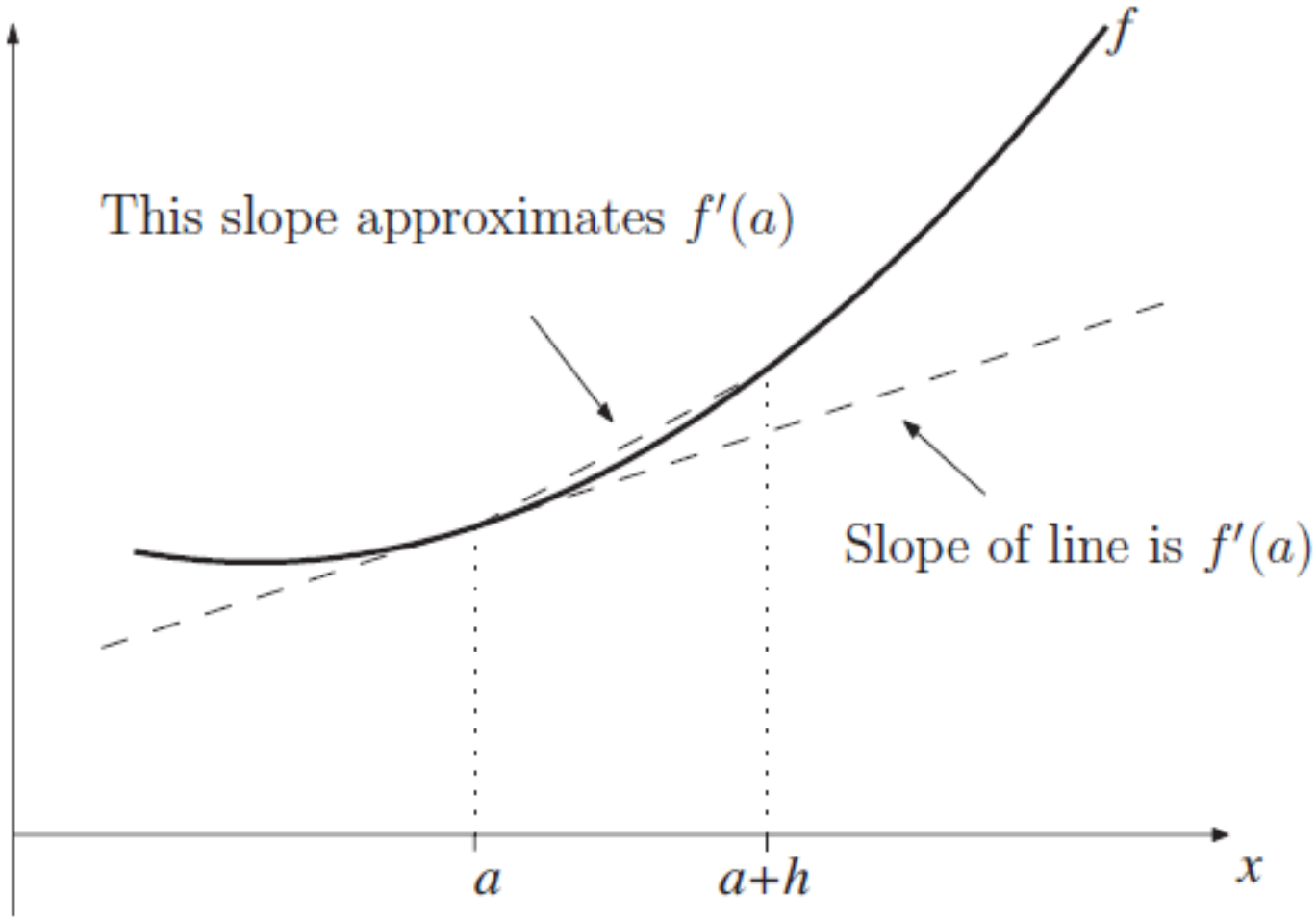
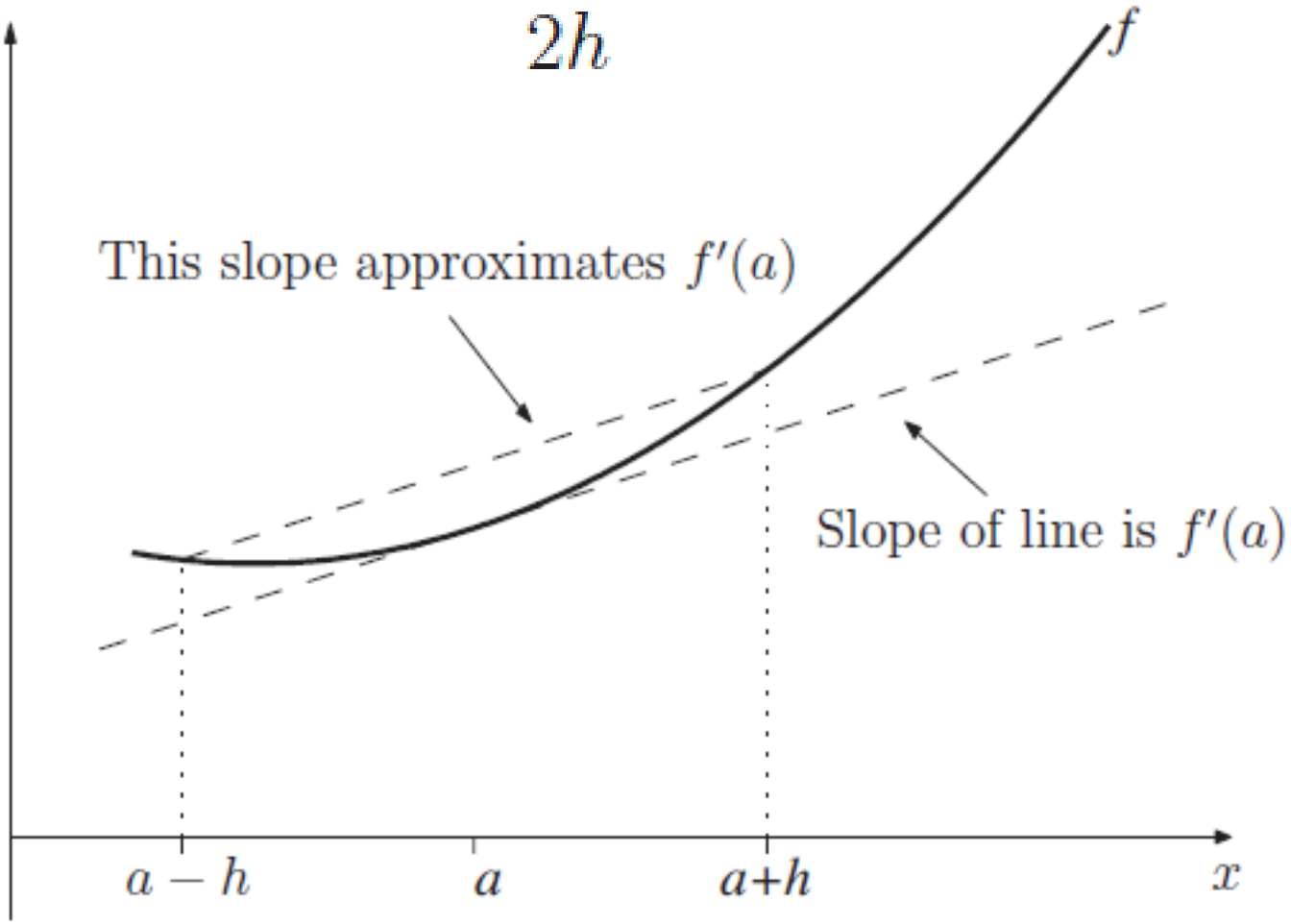
Numerical Differentiation and Integration

$$f'(a) \approx \frac{f(a) - f(a-h)}{h} \quad \text{or} \quad \frac{f(a+h) - f(a)}{h}.$$



Central Difference

$$\frac{f(x+h) - f(x-h)}{2h}$$



Definition of Derivative

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2} f''(x) + O(h^3)$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2} f''(x) + O(h^3)$$

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} + O(h^2)$$

Three Point Approximation

$$f'(x) \approx af(x) + bf(x+h) + cf(x+\alpha h)$$

- Taylor expansion and mean value theorem

$$\begin{aligned} & af(x) + bf(x+h) + cf(x+\alpha h) \\ = \; & (a+b+c)f(x) + h(b+c\alpha)f'(x) \\ & + \frac{h^2}{2}(b+c\alpha^2)f''(x) + \frac{h^3}{6}[bf'''(x_1) + c\alpha^3 f'''(x_2)] \end{aligned}$$

- Then $a+b+c=0,$

$$b+c\alpha = 1/h,$$

$$b+c\alpha^2 = 0.$$

Solution

$$a = \frac{\alpha^2 - 1}{\alpha(1-\alpha)h}, \qquad c = \frac{1}{\alpha(1-\alpha)h}.$$

$$b = -\frac{\alpha^2}{\alpha(1-\alpha)h},$$

- When α=-1, the three point approximation reduces to the center difference
- When α=2,

$$f'(x) = \frac{1}{2h}[-3f(x) + 4f(x+h) - f(x+2h)] + O(h^2).$$

Second Derivative

- By adding together the two Taylor series

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + O(h^4),$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + O(h^4),$$

$$f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O(h^2).$$

Numerical Integration

- Given a function f(x), defined on an interval [a, b], find an approximation to the integral

$$I(f) = \int_a^b f(x)\,dx$$

- Cut up $[a, b]$ into smaller sub-intervals;
- In each sub-interval, find a polynomial $p^i(x) \approx f(x)$;
- Integrate $p^i(x)$ on each sub-interval, and sum them up.

Trapezoidal Rule

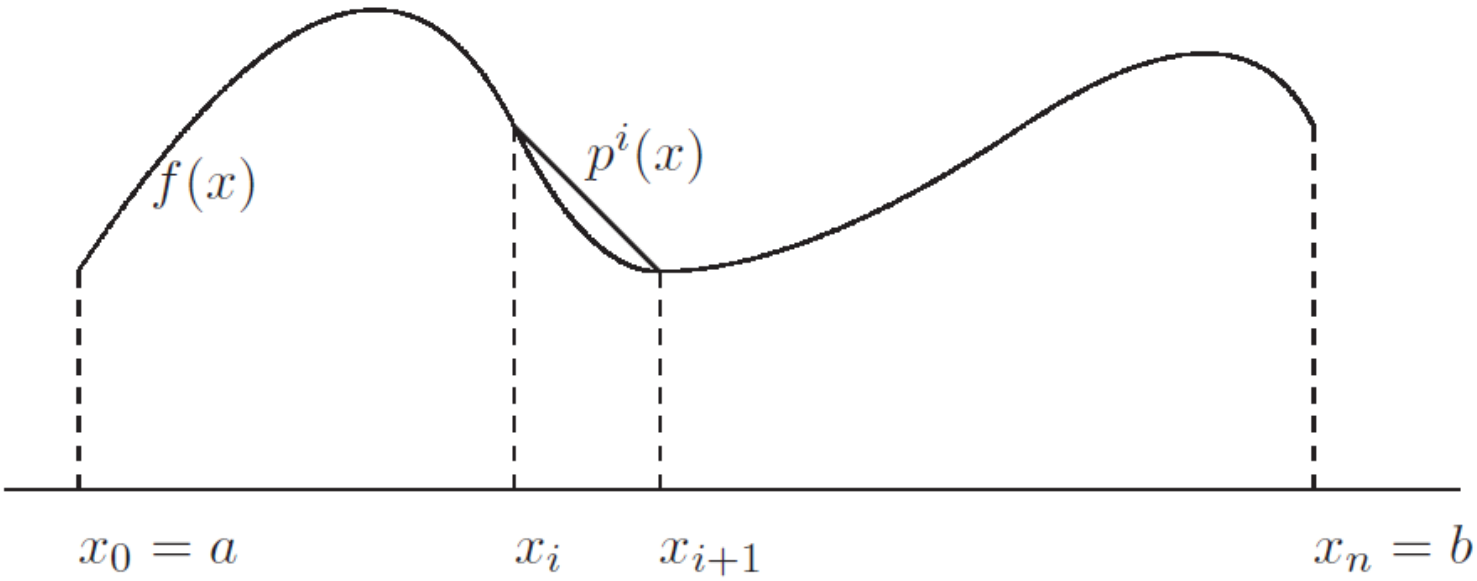
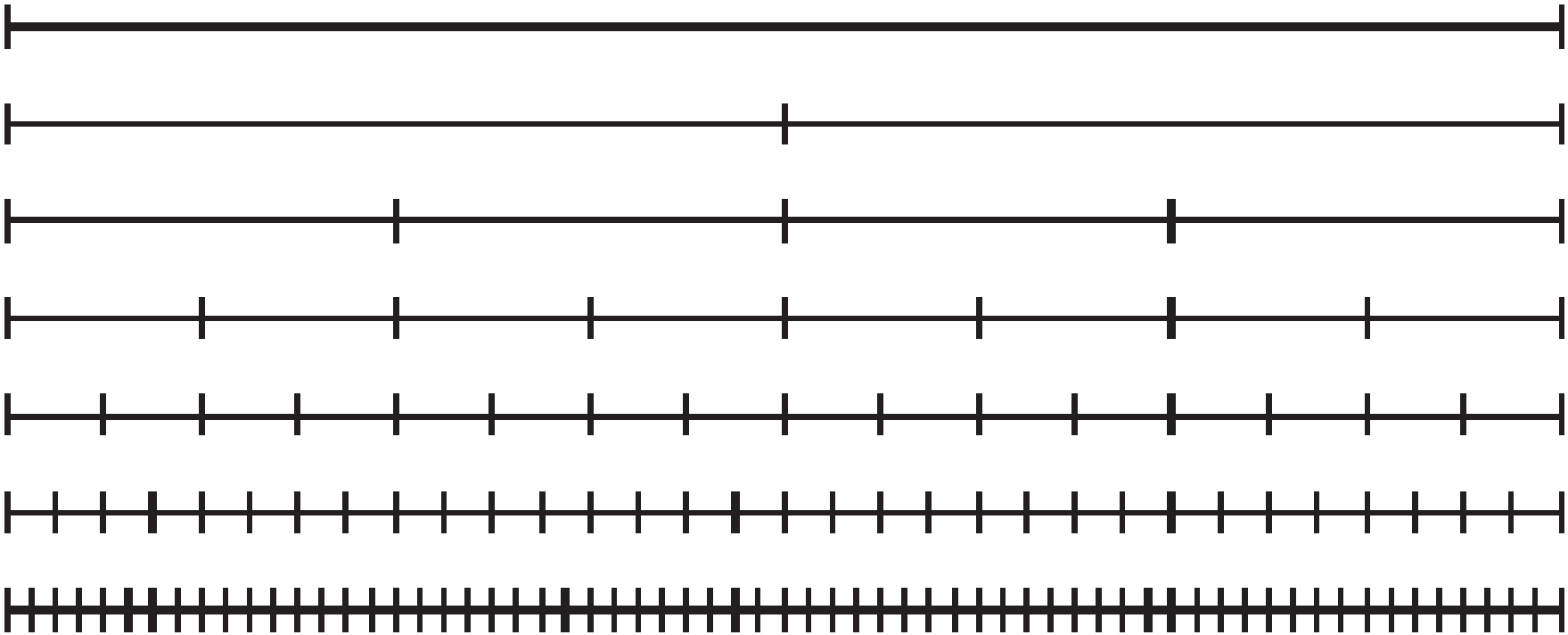


Figure 4.1: Trapezoid rule: straight line approximation in each sub-interval.

$$\int_{x_i}^{x_{i+1}} f(x)\,dx \approx \int_{x_i}^{x_{i+1}} p^i(x)\,dx = \frac{1}{2}\left(f(x_{i+1}) + f(x_i)\right)(x_{i+1} - x_i)$$

Division of Intervals



Trapezoidal Rule

- Sum up all the sub-intervals

$$\int_a^b f(x)\,dx = \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} f(x)\,dx \approx \sum_{i=0}^{n-1} \int_{x_i}^{x_{i+1}} p^i(x)\,dx$$

$$= \sum_{i=0}^{n-1} \frac{1}{2} \left(f(x_{i+1}) + f(x_i)\right)(x_{i+1} - x_i)$$

Simpson's Rule

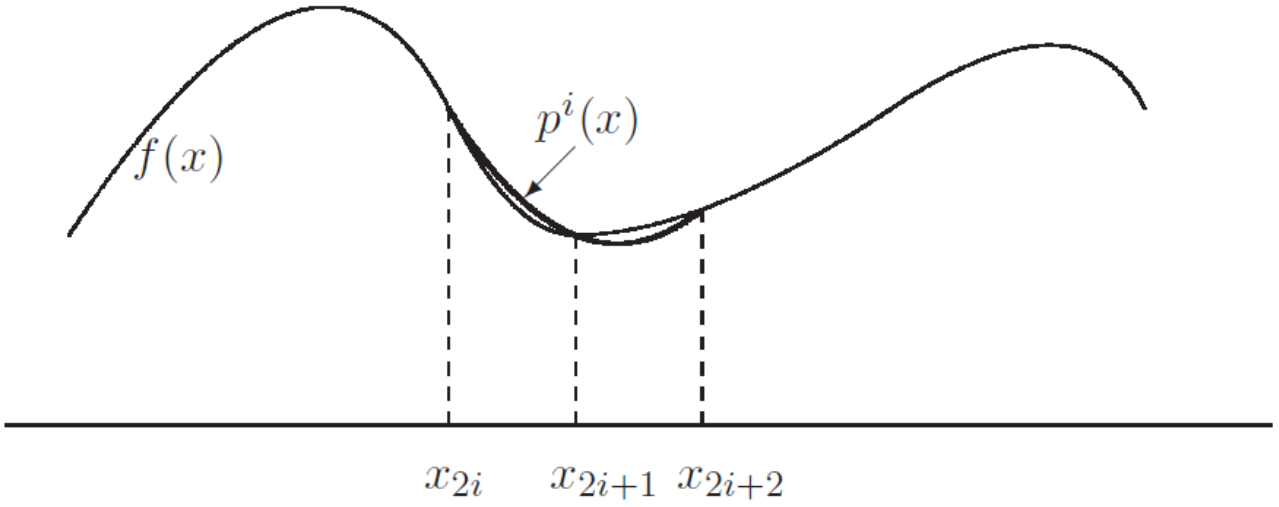


Figure 4.2: Simpson's rule: quadratic polynomial approximation (thick line) in each sub-interval.

- We cut up [a, b] into 2n equal sub-intervals

$$x_0 = a, \quad x_{2n} = b, \quad h = \frac{b-a}{2n}, \quad x_{i+1} - x_i = h$$

Lagrange Form

- Using higher order polynomials, find a 2nd order polynomial that interpolates f(x) at the points $x_{2i}, x_{2i+1}, x_{2i+2}$

$$
\begin{aligned}
p^i(x) &= f(x_{2i})\frac{(x-x_{2i+1})(x-x_{2i+2})}{(x_{2i}-x_{2i+1})(x_{2i}-x_{2i+2})} + f(x_{2i+1})\frac{(x-x_{2i})(x-x_{2i+2})}{(x_{2i+1}-x_{2i})(x_{2i+1}-x_{2i+2})} \\
&\quad + f(x_{2i+2})\frac{(x-x_{2i})(x-x_{2i+1})}{(x_{2i+2}-x_{2i})(x_{2i+2}-x_{2i+1})}
\end{aligned}
$$

$$
\begin{aligned}
p^i(x) &= \frac{1}{2h^2}f(x_{2i})(x-x_{2i+1})(x-x_{2i+2}) - \frac{1}{h^2}f(x_{2i+1})(x-x_{2i})(x-x_{2i+2}) \\
&\quad + \frac{1}{2h^2}f(x_{2i+2})(x-x_{2i})(x-x_{2i+1})
\end{aligned}
$$

Simpson's Rule

$$\begin{aligned}
\int_{x_{2i}}^{x_{2i+2}} (x - x_{2i+1})(x - x_{2i+2})\,dx &= \frac{2}{3}h^3, \\
\int_{x_{2i}}^{x_{2i+2}} (x - x_{2i})(x - x_{2i+2})\,dx &= -\frac{4}{3}h^3, \\
\int_{x_{2i}}^{x_{2i+2}} (x - x_{2i})(x - x_{2i+1})\,dx &= \frac{2}{3}h^3,
\end{aligned}$$

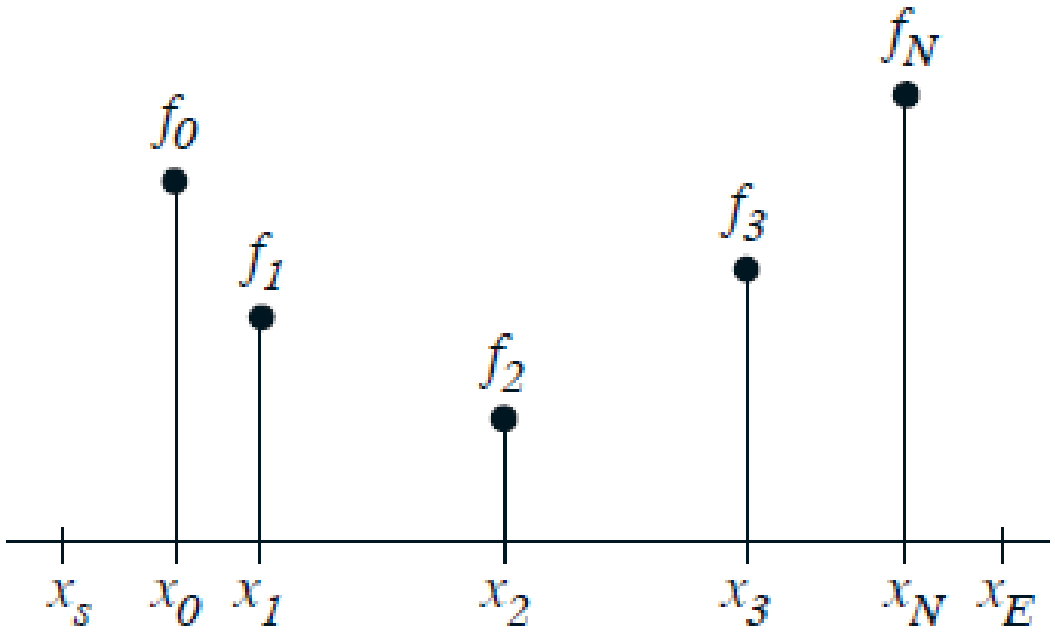
$$\int_{x_{2i}}^{x_{2i+2}} p^i(x)\,dx = \frac{h}{3}\left[f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})\right]$$

$$\int_a^b f(x)\,dx \approx \frac{h}{3}\sum_{i=0}^{n-1}\left[f(x_{2i}) + 4f(x_{2i+1}) + f(x_{2i+2})\right]$$

Gaussian Quadrature

- The form of the integration rule is

$$\int_{x_S}^{x_E} f(x)dx = [w_o f_o + w_1 f_1 + \ldots + w_N f_N] + E$$



Derive 1 Point Gauss-Quadrature

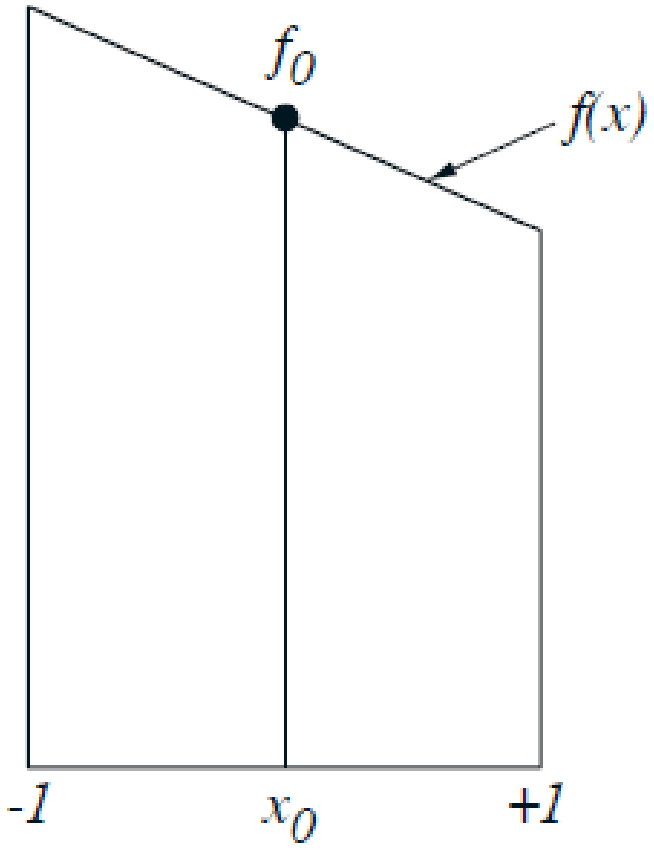
- Two unknowns w_0 and x_0 which will exactly integrate any linear function

$$\int_{-1}^{+1} f(x)dx = w_o f(x_o)$$

$$\int_{-1}^{+1} (Ax + B)dx = w_o(Ax_o + B)$$

$$\left[A\frac{x^2}{2} + Bx\right]_{-1}^{+1} = w_o(Ax_o + B) \quad \Rightarrow$$

$$A(0) + B(2) = A(x_o w_o) + B(w_o)$$



$$\begin{cases} 0 = x_o w_o \\ 2 = w_o \end{cases}$$

Derive 2 Point Gauss Quadrature Formula

- Four unknowns, we can fit a 3rd degree polynomial exactly $f(x) = Ax^3 + Bx^2 + Cx + D$



$$\int_{-1}^{+1} f(x)dx = w_o f(x_o) + w_1 f(x_1)$$

Two Points

- All polynomials of degree 3 or less will be ***exactly*** integrated with a Gauss-Legendre 2 point formula.

$$A[w_o x_o^3 + w_1 x_1^3] + B\left[w_o x_o^2 + w_1 x_1^2 - \frac{2}{3}\right]$$

$$+ C[w_o x_o + w_1 x_1] + D[w_o + w_1 - 2] = 0$$

$$w_o x_o^3 + w_1 x_1^3 = 0$$

$$w_o x_o^2 + w_1 x_1^2 - \frac{2}{3} = 0$$

$$w_o x_o + w_1 x_1 = 0$$

$$w_o + w_1 - 2 = 0$$

Solution

- 4 nonlinear equations and 4 unknowns

$$w_o = 1 \quad \text{and} \quad w_1 = 1$$

$$x_o = -\sqrt{\frac{1}{3}} \quad \text{and} \quad x_1 = +\sqrt{\frac{1}{3}}$$

$$I = \int_{-1}^{+1} f(x)dx = \sum_{i=0}^{N} w_i f_i + E$$

Gauss Legendre

N	$N+1$	x_i, $i = 0, N$	w_i	Exact for polynomials of degree
0	1	0	2	1
1	2	$-\sqrt{\frac{1}{3}}, +\sqrt{\frac{1}{3}}$	1, 1	3
2	3	-0.774597, 0, +0.774597	0.5555, 0.8889, 0.5555	5
N	$N+1$			$2N+1$

N	$N+1$	x_i, $i = 0, N$	w_i	Exact for polynomials of degree
3	4	-0.86113631 -0.33998104 0.33998104 0.86113631	0.34785485 0.65214515 0.65214515 0.34785485	7
4	5	-0.90617985 -0.53846931 0.00000000 0.53846931 0.90617985	0.23692689 0.47862867 0.56888889 0.47862867 0.23692689	9
5	6	-0.93246951 -0.66120939 -0.23861919 0.23861919 0.66120939 0.93246951	0.17132449 0.36076157 0.46791393 0.46791393 0.36076157 0.17132449	11